

MINA IBRAHIM

AI Engineer & Researcher | TensorFlow Open-Source Contributor | LLMs, Agentic AI, RAG & ML Systems

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PROFESSIONAL SUMMARY

AI Engineer and researcher building production AI systems end-to-end across large language models, agentic AI, retrieval-augmented generation, NLP, computer vision, backend APIs, and ML infrastructure. TensorFlow open-source contributor with 12 merged upstream pull requests focused on CPython free-threading, Python/C++ runtime safety, synchronization, object lifetime, pybind11, and wheel/toolchain compatibility. Experienced with TensorFlow, PyTorch, FastAPI, Linux, Bazel, CUDA, synthetic data generation, and applied AI research.

CORE COMPETENCIES

Generative AI | LLMs | Agentic AI | Multi-Agent Systems | RAG | Hybrid RAG | Knowledge Graphs | NLP | Arabic NLP | Transformers | BERT | DeBERTa | AraBERT | Machine Learning | Deep Learning | CNN | LSTM | TensorFlow | PyTorch | Python | C++ | FastAPI | REST APIs | CPython Free-Threading | Python C API | pybind11 | Concurrency | Bazel | Linux | CUDA | OpenCV | Git/GitHub | Model Evaluation | Synthetic Data

PROFESSIONAL EXPERIENCE

Library Systems & Infrastructure Engineer | Sep 2025 - Present

German - Syrian Research Foundation e.V. | Contract | Hesse, Germany | Remote

- Deploy and manage digital library and institutional repository systems for universities using Koha, DSpace, and FOLIO.
- Deliver complete platforms on university servers, including application, database, web, and network components, with domain configuration, reverse proxies, HTTPS, and public access.
- Administer system configuration, services, databases, permissions, backups, maintenance, and production reliability across Linux-based environments.
- Troubleshoot deployment, integration, authentication, networking, and server issues and support universities in maintaining reliable public research infrastructure.

OPEN-SOURCE ENGINEERING - TENSORFLOW / CPYTHON 3.14T

TensorFlow Open-Source Contributor | 2026

Independent upstream contributor to tensorflow/tensorflow

- Contributed 12 merged upstream pull requests improving TensorFlow compatibility with CPython 3.14 free-threaded execution across profiler/tracing, TFLite conversion, distributed layout, checkpoint reader, op-def/API helpers, runtime flags, DType, DTensor synchronization, and native extension safety.
- Implemented and reviewed native synchronization, Python object ownership/lifetime fixes, pybind11 mod_gil_not_used declarations, NumPy C API initialization, and cp314t ABI wheel tagging across Python/C++ boundaries.
- Built and validated an unofficial experimental TensorFlow 2.22.0 CPython 3.14t Linux x86_64 wheel and companion free-threaded gRPC wheel; the GIL remained disabled through natural TensorFlow import and concurrent TensorFlow/gRPC stress validation.

All TensorFlow contributions: <https://github.com/tensorflow/tensorflow/pulls?q=is%3Apr+author%3AMinalbrahim10>

Experimental TensorFlow 2.22.0 CPython 3.14t release: <https://github.com/Minalbrahim10/tensorflow/releases/tag/tensorflow-2.22.0-py314t-exp1>

Natural-import working snapshot: <https://github.com/Minalbrahim10/tensorflow/releases/tag/py314t-natural-import-pass-20260819>

SELECTED ENGINEERING PROJECTS

Agentic-Nexus - Multi-Agent AI System

- Designed a composable multi-agent architecture with orchestration, Hybrid RAG, FAISS retrieval, knowledge graphs, tool use, coding workflows, and automated evaluation.

GitHub: <https://github.com/Minalbrahim10/Agentic-Nexus>

AI Chatbot - Transformer LLM Built from Scratch

- Designed the transformer architecture, attention mechanism, tokenizer, training pipeline, evaluation workflow, and FastAPI real-time inference layer; generated and curated a 100,000+ sample Q&A training dataset.

AI-Powered Journal Management System

- Built a full-stack academic AI platform for submission screening, classification, reviewer matching, citation analysis, plagiarism signals, summarization, and editorial workflows using FastAPI and NLP models.

Hotel 1000 - Hotel Management System

- Architected and delivered a production web platform covering reservations, guest profiling, operational analytics, backend APIs, database design, frontend integration, and deployment.

Live project: <https://hotel1000.me>

Multimodal Defect Detection

- Developed TensorFlow/OpenCV pipelines and custom CNN/Transformer models for defect detection across image, text, and audio modalities.

RESEARCH & PUBLICATIONS

- Auditing Internal Consistency in Webometrics University Rankings: A Bidirectional Historical and Peer-Calibrated Approach
https://www.researchgate.net/publication/412783335_Auditing_Internal_Consistency_in_Webometrics_University_Rankings_A_Bidirectional_Historical_and_Peer-Calibrated_Approach
- Automatic Citation Style Classification Using Multilingual BERT and Transformer-Based Architecture
https://www.researchgate.net/publication/396353830_Automatic_Citation_Style_Classification_Using_Multilingual_BERT_and_Transformer-Based_Architecture
- Large-Scale Hybrid Dialogue Data Processing for Transformer-Based Generative Chatbots Using Pretrained DeBERTa Embeddings
https://www.researchgate.net/publication/395337384_Large-Scale_Hybrid_Dialogue_Data_Processing_for_Transformer-Based_Generative_Chatbots_Using_Pretrained_DeBERTa_Embeddings
- Detecting Machine-Generated Arabic Text: AraBERT-LSTM for Trustworthy Low-Resource NLP
https://www.researchgate.net/publication/395334508_Detecting_Machine-Generated_Arabic_Text_AraBERT-LSTM_for_Trustworthy_Low-Resource_NLP
- KM-Chat: A Large-Scale Synthetic Question-Answer Dataset for Open-Domain Conversational AI
https://www.researchgate.net/publication/394373244_KM-Chat_A_Large-Scale_Synthetic_Question-Answer_Dataset_for_Open-Domain_Conversational_AI

TENSORFLOW UPSTREAM EVIDENCE

#125938 - Make debug events writers safe for free-threaded Python | <https://github.com/tensorflow/tensorflow/pull/125938>
#125648 - Make profiler and tracing bindings safe for free-threaded Python | <https://github.com/tensorflow/tensorflow/pull/125648>
#125646 - Make TFLite conversion bindings safe for free-threaded Python | <https://github.com/tensorflow/tensorflow/pull/125646>
#125643 - Make distributed layout bindings safe for free-threaded Python | <https://github.com/tensorflow/tensorflow/pull/125643>
#125642 - Make Python API helper bindings safe for free-threaded Python | <https://github.com/tensorflow/tensorflow/pull/125642>
#125640 - Make op-def bindings safe for free-threaded Python | <https://github.com/tensorflow/tensorflow/pull/125640>
#125636 - Make checkpoint reader safe for free-threaded Python | <https://github.com/tensorflow/tensorflow/pull/125636>
#125581 - Lock DTensor collective key cache before lookup | <https://github.com/tensorflow/tensorflow/pull/125581>
#125569 - Mark DType extension as free-threading safe | <https://github.com/tensorflow/tensorflow/pull/125569>
#125566 - Mark runtime flag extensions as free-threading safe | <https://github.com/tensorflow/tensorflow/pull/125566>
#125564 - Mark small Python extensions as free-threading safe | <https://github.com/tensorflow/tensorflow/pull/125564>
#125493 - Set free-threaded Python wheel ABI tag | <https://github.com/tensorflow/tensorflow/pull/125493>
Related upstream gRPC PR #43278 (open): <https://github.com/grpc/grpc/pull/43278>
Related rules_ml_toolchain PR #302 (open): https://github.com/google/ml-infra/rules_ml_toolchain/pull/302

EDUCATION

Bachelor of Science - Information Technology & Artificial Intelligence | Oct 2019 - Jul 2024

Arab International University | Damascus, Syria

- Major focus: artificial intelligence, machine learning, neural networks, computer vision, NLP, and software systems.

CERTIFICATION & LANGUAGES

Certification: Introduction to Deep Learning with OpenCV

Languages: Arabic - Native | English - Professional Working Proficiency

PROFESSIONAL LINKS

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